Linear Systems Theory

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Module 2

Lecture 3

Math Preliminaries: Linear Algebra 2

Span of a Vector Space

Linear Combination of Vectors

Any combination of a number of vectors by means of vector addition and scalar multiplication of the vectors forms a linear combination of the vectors.

- Eg. $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $\mathbf{b} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$; $\mathbf{c} = 2\mathbf{a} + 3\mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

  $\mathbf{c}$ is a linear combination of the vectors $\mathbf{a}$ and $\mathbf{b}$.

- In general, a linear combination of n vectors is given as:

$$\mathbf{v} = \mathrm{k}_1\mathbf{v}_1 + \mathrm{k}_2\mathbf{v}_2 + \ldots + \mathrm{k}_n\mathbf{v}_n$$

  where $\mathrm{k}_1, \mathrm{k}_2, \ldots, \mathrm{k}_n \in \mathbb{R}$ or $\mathbb{F}$.

Linear Independence of Vectors

Let $(V, \mathbb{F})$ be a vector space and let $\{\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n\}$ be a set of vectors in V ($\mathbf{v}_i \in V$).

- The set of vectors, $\{\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n\}$, are called linearly independent, iff $\exists$ scalars, $\mathrm{k}_i \in \mathbb{F},\ i = 1, ..., n$ such that

$$\mathrm{k}_1\mathbf{v}_1 + \mathrm{k}_2\mathbf{v}_2 + \ldots + \mathrm{k}_n\mathbf{v}_n = \mathbf{0}, \quad \Rightarrow \mathrm{k}_1 = \mathrm{k}_2 = ... = \mathrm{k}_n = \mathbf{0}$$

  In other words, the linear combination of vectors results in a zero vector, iff all the scalar multipliers are zero.

- The set of vectors is said to be linearly dependent iff $\exists$ scalars, $\mathrm{k}_i \in \mathbb{F},\ i = 1, ..., n$, **not all zero** and $\mathrm{k}_1\mathbf{v}_1 + \mathrm{k}_2\mathbf{v}_2 + \ldots + \mathrm{k}_n\mathbf{v}_n = \mathbf{0}$

Span

Let V be a vector space and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a subset of V. Then S *spans* V if every vector $\mathbf{v}$ in V can be written as a linear combination of vectors in S i.e.

$$\mathbf{v} \;=\; c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n.$$

where $\mathbf{v}_i \in S$, $\mathbf{v} \in V$ and $c_i \in \mathbb{R}$ for $i = 1, \ldots, n$

- $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ spans the full two dimensional space $\mathbb{R}^2$.
- Does $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ span the space $\mathbb{R}^2$?
- What about the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$, do they span the space $\mathbb{R}^2$?

Basis of a Vector Space

Basis of a Vector Space

Let $(V, \mathbb{F})$ be a (linear) vector space. A set of vectors $\{\mathbf{b}_1, ..., \mathbf{b}_n\}$ is called a basis of V if

$$\text{span}\,\{\mathbf{b}_1, \mathbf{b}_2, ..., \mathbf{b}_n\} = V$$

and any vector $\mathbf{v} \in V$ can be written as a linear combination of these basis vectors as

$$\mathbf{v} = \mathrm{k}_1\mathbf{b}_1 + \mathrm{k}_2\mathbf{b}_2 + ... + \mathrm{k}_n\mathbf{b}_n, \ \forall\, \mathbf{v} \in V$$

where the set of vectors $\{\mathbf{b}_1, \mathbf{b}_2, ..., \mathbf{b}_n\}$ are linearly independent.

- $\mathrm{k} = [\mathrm{k}_1,\ \mathrm{k}_2, ..., \mathrm{k}_n]^T \in \mathbb{F}$ is called the *coordinate representation, or coordinate vector* of $\mathbf{v}$ with respect to the basis $\{\mathbf{b}_1, .., \mathbf{b}_n\}$.
- The k_i's are called the coordinates of $\mathbf{v}$ with respect to the basis $\{\mathbf{b}_i\}$.
- The number of basis vectors is called the dimension of the vector space.

Basis of a Vector Space

Question: Is the representation of a vector unique with respect to a given basis? In other words, can there be another set of coordinates $\{k_1^{'}, \ldots, k_n^{'}\}$ that represent the same vector $\mathbf{v}$ with the given basis $\{\mathbf{b}_1, ..., \mathbf{b}_n\}$?

Basis of a Vector Space

- The standard basis for $\mathbb{R}^3$ are

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

- The basis is not unique: For example

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}; \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

is also a basis for $\mathbb{R}^3$

Orthonormal Basis

- Two vectors are orthonormal if they are orthogonal and of unit length.
- Vector set V is an orthonormal basis if each pair of vectors in V are orthonormal.
- Eq. $V = \left\{ \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix} \right\}$ form an orthonormal basis for $\mathbb{R}^2$.
- Eq. $V = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ form an orthonormal basis for $\mathbb{R}^2$

Advantages of Orthonormal Basis

- Convenient basis because the coordinates of a vector can be easily calculated.
- $\mathbf{v} \in \mathbb{R}^n$ and basis of $\mathbb{R}^n = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$, then $\mathrm{k}_i = \mathbf{v}.\mathbf{b}_i \ \forall \ i = \{1, \ldots, n\}$.
- Standard basis is the most popularly used orthonormal basis because $\mathrm{k}_i = v_i$

Exercise 1

What is the coordinate representation of vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ w.r.t. the orthonormal basis $\left\{ \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix} \right\}$?

Vector Subspace

Vector Subspace

- Let $(V, \mathbb{F})$ be a (linear) vector space and let U be a subset of V, $U \subset V$.
- Let $(U, \mathbb{F})$ is called a subspace of $(V, \mathbb{F})$ if $(U, \mathbb{F})$ is itself a subspace.
- How to check if U is a subspace?
  - U contains zero vector i.e. $\mathbf{0} \in U$.
  - Closure under scalar multiplication i,e,. if $\mathbf{a} \in U$; $c \in \mathbb{R}$, then $c\,\mathbf{a} \in U$.
  - Closer under addition i,e,. if $\mathbf{a}, \mathbf{b} \in U$, then $\mathbf{a} + \mathbf{b} \in U$.
- $\dim U \leq \dim V$
- Examples of subspaces:
  - $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is a 0-dimensional subspace of $\mathbb{R}^2$.
  - $\text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a 1-dimensional subspace of $\mathbb{R}^2$.
  - A plane in $\mathbb{R}^3$ is a subspace.

Question: If U_1 and U_2 are subspaces is $U_1 \cup U_2$ a subspace?

Exercise 1

Is $U_1 \cap U_2$ a subspace?

Overview

Summary: Mod 2 Lecture 3

- Span of a vector space
- Basis of a vector space
- Vector subspace

Contents: Mod 2 Lecture 4

- Linear maps / transformations
- Kernel and Image of linear maps
- Matrices as Linear Maps